

Radiotherapy for prostate cancer

Radiotherapy uses high-energy rays to treat cancer. It is often given from outside the body (external beam radiotherapy). It can also be given from inside the body (brachytherapy).

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What is radiotherapy?

[Radiotherapy](#) uses high-energy rays to treat cancer. It destroys cancer cells in the area where the radiotherapy is given. The aim of radiotherapy for [prostate cancer](#) is to try to cure the cancer, or control it for many years. Doctors call this radical radiotherapy. They try to make sure radiotherapy causes as little harm as possible to healthy tissue and nearby areas, such as the bladder and bowel.

Radiotherapy for prostate cancer can be given in 2 ways:

- [External beam radiotherapy](#) is given from outside the body (externally) using a radiotherapy machine called a linear accelerator (LINAC).
- Internal radiotherapy is when a radioactive material is placed inside the body. It is called [brachytherapy](#).

This information is about external beam radiotherapy. We have separate information about [brachytherapy for prostate cancer](#).

You might have radiotherapy with [hormonal therapy](#). Hormonal therapy can shrink the cancer, which helps make radiotherapy more effective. Your cancer doctor may advise you to have hormonal therapy for 3 to 6 months before, during or after radiotherapy.

You may continue with hormonal therapy for up to 3 years. Some people may also be given another hormonal therapy called [abiraterone](#) for 2 years.

If you are having radiotherapy, you will see a clinical oncologist. A clinical oncologist is a cancer doctor who uses radiotherapy, chemotherapy and other cancer drugs to treat cancer.

You will also meet a specialist therapeutic radiographer. Therapeutic radiographers are experts in radiotherapy and are specially trained in cancer treatment. They can also give you support, advice and information about your radiotherapy.

Smoking

If you smoke, it is important to try to stop. [Stopping smoking](#) can make radiotherapy work better. It also reduces the [side effects](#) of treatment.

Smoking also increases the risk of bone thinning. This is called osteoporosis. Smoking is also a major risk factor for smoking-related cancers and [heart disease](#).

It can be difficult to stop smoking, but you can get support. Your doctor or nurse can give you advice. There are also stop smoking services to help.

Having radiotherapy for prostate cancer

You have [external beam radiotherapy](#) as an outpatient in the radiotherapy department. Radiotherapy is given using a machine called a linear accelerator. This is often called a LINAC.

You usually have radiotherapy as a series of short, daily treatments (fractions). Each treatment lasts around 10 minutes. But you will be in the radiotherapy department for longer. The treatments are given from Monday to Friday, with a rest at the weekend. Radiotherapy is not painful, but you need to lie still while you have it.

Depending on the type of radiotherapy, treatment fractions are usually given over 1 to 4 weeks. Your cancer doctor or radiographer will explain your treatment plan.

The radiotherapy does not make you radioactive. It is safe for you to be with other people during external radiotherapy, including children.

In this video, Tony and his husband Dave talk about their experience of prostate radiotherapy, and radiographer Claire answers common questions about treatment and what to expect.



Types of radiotherapy for prostate cancer

There are different ways of giving external beam radiotherapy. The type you have can depend on the [risk group](#) of the cancer and symptoms. These all target the prostate cancer very precisely. This aims to treat the cancer while protecting healthy tissue. This can help reduce side effects and late effects.

Intensity modulated radiotherapy (IMRT)



IMRT uses computers to calculate and deliver radiation directly to the cancer from different angles. It shapes the radiation beams to the size of the tumour. The strength (intensity) of the dose can be changed depending on the tissue. This means delivering a higher radiation dose to the cancer while giving lower doses to healthy tissue.

Volumetric modulated arc therapy (VMAT)



The radiation dose can be changed even more accurately during treatment. The machine rotates around you and delivers radiotherapy beams in continuous arcs (curves) precisely to the cancer. VMAT can be given in shorter treatment sessions.

Stereotactic ablative radiotherapy (SABR)



[SABR](#) allows large doses of radiotherapy to be given very precisely to small areas. The beams are directed from different angles that meet at the tumour. It involves giving larger doses of radiation to the prostate. This means you have treatment fractions over a few days, instead of weeks.

Planning radiotherapy for prostate cancer

Your cancer doctor carefully plans your radiotherapy to make it as effective as possible, while keeping side effects to a minimum.

Having your planning scan

During the [planning visit](#), you will have a [CT scan](#). The scan is done to help your radiotherapy team plan the dose and area of your treatment. The hospital will send you information if you need to prepare for your scan.

Before your scan, you may need to have a special diet or take medicine to empty your bowel. You may also need to drink water to fill your bladder. At some hospitals, you may be asked to go to the toilet so your bladder is empty. Doing these things helps get very clear CT pictures for planning your treatment.

You may have a small amount of liquid passed into your rectum to empty your bowel. This is called an enema. It helps you empty your bowel fully when you go to the toilet. You can usually do this yourself. If you need help, speak to your radiotherapy team.

During the scan, you need to lie still in the same position you will be in for your radiotherapy.

Tattoo marks

A therapeutic radiographer will give you your treatment. They may make some permanent marks (tattoos) the size of a pinpoint on your skin. These are used to make sure you are in the correct position for your treatment. They also show where the beams will be directed. This is only done with your permission. It may be a little uncomfortable. If you are worried about this, talk to the radiographer.

Other treatment planning

Your doctor may talk to you about having some of the following or you may hear about these:

Fiducial markers



Sometimes you may have tiny, metallic grains passed into your prostate using an ultrasound probe. These are called fiducial markers. They help the radiographer see the position of the prostate before each session. This may help reduce side effects and possible damage to nearby areas, such as the bladder or bowel.

Image guided radiotherapy (IGRT)



The prostate can change position depending on the size of the bladder or bowel. With IGRT, the radiographers scan the prostate area before and sometimes during each session of radiotherapy. The scan pictures show the size and shape of the prostate cancer and the position of the bowel and bladder. The radiographers compare these pictures to the planning scan to check the position and the treatment area. They can then ensure the prostate is accurately targeted before each treatment. This makes the radiotherapy very precise.

Rectal spacers



A small amount of liquid gel, or an inflatable biodegradable balloon, is put into the space between the prostate and rectum before treatment. It moves the rectum away from the prostate and reduces the amount of radiation reaching the rectum. This can help reduce side effects to the rectum.

Rectal spacers are not available in all hospitals. They are not needed, or are not suitable, for everyone.

Your cancer doctor can tell you more about what is involved. You can ask them whether rectal spacers are available and whether they are suitable for you. They may be available through private healthcare.

Having a treatment session

The radiographer will explain what will happen. At the start of each treatment session, they make sure you are in the correct position on the couch and that you are comfortable.

When everything is ready, they leave the room and give you your treatment. This only takes a few minutes. You will be on your own during the treatment but can talk to the radiographers through an intercom or signal to them during the treatment. They can see and hear you from the next room.

During treatment, the radiotherapy machine may automatically stop and move into a new position. This is so the radiotherapy can be given from different directions.

Side effects of radiotherapy for prostate cancer

Side effects usually build up slowly after you start treatment. They may continue to get worse for a couple of weeks after treatment. But after this, most side effects improve gradually over the next few weeks.

Your [radiotherapy team](#) will talk to you about this. They will explain what to expect and give you advice on what you can do to manage side effects.

Always tell them about your side effects rather than trying to treat them yourself. There are usually things they can do to help. We list the common side effects here, but you may not get all of these. We have more detailed information about [pelvic radiotherapy](#).

Bladder side effects during treatment

Radiotherapy can also cause inflammation of the bladder. This is called radiation cystitis. You may:

- feel you want to pass urine (pee) more often (frequency)
- have a burning feeling when you pee
- be unable to wait to empty your bladder (urgency)
- have difficulty in starting to pee.

Your cancer doctor or radiographer can prescribe medicines to help. Drinking 2 to 3 litres (3½ to 5½ pints) of fluids a day can help. Avoid drinks containing caffeine and alcohol. These side effects usually disappear slowly a few weeks after treatment has finished.

Some people may have difficulty passing urine, but this is rare. If you are having problems, you may need to have a tube put into the bladder to drain urine. This is called a urinary catheter.

Side effects can be worse if you already had bladder problems before starting radiotherapy. If you already had bladder problems before treatment, you may find them start to improve after radiotherapy finishes.

Bowel side effects during treatment

Radiotherapy to the prostate can irritate the lower bowel (rectum). You may get diarrhoea, wind and cramping pains in your tummy (abdomen). You may feel your bowel is not completely empty after going to the toilet. You may notice blood or mucus when you go to the toilet. It is important to tell your radiotherapy team about bowel side effects.

If you have diarrhoea, drink at least 2 to 3 litres (3½ to 5½ pints) of fluids a day. Avoid caffeine and alcohol. During treatment, your radiographer may advise you to make some changes to your diet, such as eating less fibre. These side effects usually improve by 6 weeks after finishing treatment.

Your cancer doctor, specialist nurse or pharmacist may give you anti-diarrhoea drugs to take at home. It is important to follow their advice about taking the drugs.

Symptoms can often be managed with medication and changes to your diet. If problems do not improve, you can ask to be referred to a late effects specialist or a bowel specialist – this may be a gastroenterologist or bowel surgeon. We have more information about [bowel problems after pelvic radiotherapy](#).

Tiredness

Radiotherapy often makes people feel tired. Hormonal therapy can add to the tiredness. [Tiredness \(fatigue\)](#) may get worse as treatment goes on. There are things you can do to help:

- Get plenty of rest, but keep your daily routine if you feel able to.
-

Do some regular exercise, such as going for a walk.

- Eat a healthy diet and drink plenty of fluids.
- Ask for help with everyday tasks, if you have friends or family members who can support you.

After treatment finishes, tiredness should improve. If it does not get better after a few weeks, tell your cancer doctor or specialist nurse.

Effects on the skin

Less commonly, the skin in the treated area may change colour. If you have white skin, it may become red. If you have black or brown skin, it may look darker.

The skin may also become dry or flaky, and feel itchy or tight. Sometimes the skin around the anus and scrotum becomes moist and sore. Your radiographer will give you advice. Your cancer doctor or radiographer can prescribe a cream or dressings and painkillers if you need them.

Your pubic hair may fall out. It usually starts to grow back a few weeks after you have finished treatment. It may be thinner than before.

Erection problems

Radiotherapy for prostate cancer can cause problems getting or keeping an erection. This is called [erectile dysfunction \(ED\)](#). ED may not happen straight away, but it can develop slowly over 2 to 5 years. About 30 to 45 out of 100 people (30% to 45%) who do not have any problems before radiotherapy develop ED after treatment.

You may have a higher risk of ED if:

- you are older
- you already had ED before treatment
- you have other medical problems that also cause ED, such as diabetes or heart disease
- you are taking hormonal therapy, which also causes ED and affects your desire to have sex (libido).

Ask your [radiotherapy team](#) about your risk of ED. If you develop ED, there are different treatments that can help.

After radiotherapy and brachytherapy, orgasms might feel different and you might ejaculate little or no semen. Hormonal therapy can also make your penis appear shorter in length. Your cancer team can talk to you about ways of preventing or helping this.

Fertility

Radiotherapy to the prostate may cause [infertility](#). This means you will no longer be able to [to get someone pregnant](#). If you find this difficult to cope with or are worried, talk to your cancer doctor.

You may be able to [store sperm](#) before treatment starts.

Late effects of radiotherapy for prostate cancer

Side effects that do not improve or happen months to years after radiotherapy are called long term or late effects. Improved ways of giving radiotherapy are reducing the risk of late effects, particularly on the bowel. Your [radiotherapy team](#) will explain these to you.

Late bladder effects

Bladder side effects you get with treatment (frequency and urgency) may not completely go away, or they may develop later. Sometimes radiotherapy can affect how well your bladder can hold urine.

The bladder lining may bleed easily, causing blood in your urine. This is called haematuria. You may get leakage of small amounts of urine (urinary incontinence). But this is rare. We have more information about how [bladder problems may be treated](#).

Late bowel effects

With newer radiotherapy methods, late effects to the bowel are less common. Late effects may be similar to the immediate side effects. You may feel you need to rush urgently to go to the toilet. Or, rarely, you might have some leakage or soiling (bowel incontinence). Sometimes blood vessels in the bowel lining become more fragile and bleed. If you notice any bleeding, always tell your doctor so they can check it.

Symptoms can often be managed with medication and changes to your diet. If problems do not improve, you can ask to be referred to a late effects specialist or a bowel specialist. This may be a gastroenterologist or bowel surgeon. We have more information about [bowel problems after pelvic radiotherapy](#).

Benefits and disadvantages of external beam radiotherapy for early prostate cancer

Radiotherapy and a radical prostatectomy are both treatments that may cure [early prostate cancer](#). They are equally effective in treating the cancer.

To choose the treatment that is best for you, it can help to look at the benefits and disadvantages of each one. You can then make your decision with your healthcare team.

Benefits

- You do not need [surgery](#) or a general anaesthetic, which may have risks if you have other health conditions.
- You can usually keep doing most of your daily routine.
- Urinary problems may happen less often than after a prostatectomy.

Disadvantages

- Depending on your treatment plan, you go to the radiotherapy department for treatments for 1 to 4 weeks.
- Bowel side effects may be worse in the short term than with a prostatectomy.
- It may be some time before you and your cancer doctor know if treatment has been successful.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Date reviewed

Reviewed: 01 October 2025 | Next review: 01 October 2028

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We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

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